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324631 (24)

BE (6th Semester)

Examination, Nov-Dec., 2014

Branch : Elect.

FIBRE OPTICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : All questions are compulsory. Part (a) of each question is compulsory. Attempt any two parts from part (b), (c) and (d).

Q. 1. (a) Define acceptance angle of an optical fiber. 2

(b) An step index optical fiber has a numerical aperture of 0.2 and cladding refractive index 1.59. Determine : 7

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(i) The acceptance angle of the fiber in water which have refractive index 1.33.

(ii) The critical angle of core and cladding interface.

(iii) The number of mode transmitted at 1300 nm wavelength 25 μm core radius.

(c) With simple ray theory describe the mechanism of transmission of light with in optical fiber. 7

(d) Write short note on the following :

(i) Mode of propagation 4

(ii) Index profile 3

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Q. 2. (a) Write down two advantages and disadvantages of LEDs. 2

(b) What is a photodetector? With a neat sketch explain the working of PIN photodetector. Also write advantages and disadvantages and application. 7

(c) Write short notes on the following :

(i) Response time. 4

(ii) Avalanche multiplication noise. 3

(d) When a 3 × 10" photon each with wave length of 0.85 μm are incident on a photodiode on average 1.2 × 10" electrons

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are collected at the terminal of the device. Determine the quantum efficiency and responsivity of photodiode at 0.85 μm. 7

Q. 3. (a) What is chromatic dispersion in optical fiber. 2

(b) A multi-mode step-index-fiber has numerical aperture of 0.3 and core refractive index 1.51. The material dispersion parameter for the fiber 270 PS nm^{-1} km^{-1} which make material dispersion totally dominating intermodal dispersion mechanism. Find :

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(i) The total R.M.S. pulse broadening per km when the fiber used with LED source of RMS spectral width 60 nm. 4

(ii) The corresponding bandwidth length of the fiber. 3

(c) What is intermodal dispersion distortion? Write down the expression of maximum pulse broadening from intermodal dispersion. 7

(d) Explain the design optimization of single mode fiber. 7

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Q. 4. (a) Define Bit Error Rate (BER). 2

(b) What do you understand by line coding What are the NRZ, RZ and BRZ signaling in optical fiber transmission link. 7

(c) Discuss with block diagram a coherent optical fiber communication system. 7

(d) Write short notes on the following :

(i) Rise time budget 4

(ii) Link power budget 3

Q. 5. (a) What do you understand by ring topology. 2

(b) Write short notes on the following :

(i) FDDI 4

(ii) SONET 3

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(c) Explain EFDA with block diagram? 7

(d) What do you understand by wavelength division multiplexing and also write down main features of WDM. 7

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